

Ordinal Numbers

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1 Well-Ordered Sets

Let J, K be well-ordered sets. J, K are said to be order isomorphic if there exists a bijection $h : J \rightarrow K$ such that if $j_1 < j_2$, then $h(j_1) < h(j_2)$.

Order isomorphic is symmetric. Suppose $k_1 < k_2$. Since h is bijective, there exist $j_1, j_2 \in J$ such that $h(j_1) = k_1$ and $h(j_2) = k_2$. Sps FTSOC $j_1 \geq j_2$. Then $h(j_1) \geq h(j_2)$, or $k_1 \geq k_2$, cont. Hence, $j_1 < j_2$ and $h^{-1}(k_1) < h^{-1}(k_2)$, so $h^{-1} : K \rightarrow J$ is an order isomorphism.

Let J, K, L be well-ordered such that $J \xrightarrow{h} K, K \xrightarrow{k} L$. Then $J \cong L$. $k \circ h : J \rightarrow L$ is bijective.

$$j_1 < j_2 \rightarrow h(j_1) < h(j_2) \rightarrow k(h(j_1)) < k(h(j_2)).$$

Let $J \xrightarrow{h} K$ be well-ordered. Then $h((a, b)) = (h(a), h(b))$. There are analogous forms for intervals with the shapes $[], [], []$. Since h is order-preserving, $h((a, b)) \subset (h(a), h(b))$. h^{-1} is also order-preserving.

$$h^{-1}((h(a), h(b))) \subset (a, b) \rightarrow (h(a), h(b)) \subset h((a, b)).$$

Then $h((a, b)) = (h(a), h(b))$.

Let $h : J \rightarrow J$ be an automorphism. Then $h = \text{id}_J$. Suppose FTSOC there exists j such that $h(j) > j$. Let j_0 be minimal. Then $h(j_0) > j_0 > h^{-1}(j_0)$, cont min. Hence, $h(j) \leq j$ for all $j \in J$. Similarly, $h^{-1}(j) \leq j$, so $j \leq h(j)$, so $h(j) = j$.

If $h, h' : J \rightarrow K$ are isomorphisms, then $h = h'$. $h'^{-1} \circ h : J \rightarrow J$ is an automorphism of J , and equals id_J , so $h = h'$.

“Proto” transfinite induction. Let J be well-ordered, $J_0 \subset J$ s.t. J_0 has the two properties:

(1) If $\alpha \in J_0$ and α has an immediate successor α^+ in J , then $\alpha^+ \in J_0$.

(2) If $\alpha \in J$ has no immediate predecessor and for all $\beta < \alpha$, we have $\beta \in J_0$, then $\alpha \in J_0$.

Then $J_0 = J$. Suppose $J_0 \subsetneq J$. Let $\alpha \in J - J_0$ be minimal. If α has no immediate predecessor, then for all $\beta < \alpha$, we have $\beta \in J_0$ (otherwise cont min). Then by (2), we have $\alpha \in J_0$. Else, α has an immediate predecessor $\alpha^- \in J_0$. Since $\alpha^{-+} = \alpha$, we have $\alpha \in J_0$, cont.

Let J be well-ordered and $j \in J$. Then $J \not\cong J(j)$. Suppose $h : J \rightarrow J(j)$ is an isomorphism. Then $h(j) < j$. Construct $a_n = h^{(n)}(j)$, so $\{a_n\}$ is an infinite descending chain, cont.

If J, K are well-ordered, then one of the following is true:

- (1) $J \cong K$.
- (2) $J \cong K(k)$ for some $k \in K$.
- (3) $K \cong J(j)$ for some $j \in J$.

Define $F = \{(j, k) \in J \times K : J(j) \cong K(k)\}$. Suppose $(j_1, k), (j_2, k) \in F$. Then $J(j_1) \cong K(k) \cong J(j_2)$. If $j_1 \neq j_2$ WLOG $j_1 < j_2$. Then $J(j_1) = J(j_2)(j_1)$, so $J(j_1) \not\cong J(j_2)$. Hence, $j_1 = j_2$. Similarly, if $(j, k_1), (j, k_2) \in F$, then $k_1 = k_2$. F is a well-defined bijective function (not necessarily over $J \times K$).

First, suppose there exist j, k such that $j \notin \text{Dom}(F), k \notin \text{Rng}(F)$. Take j, k to be minimal. Suppose $(j', k') \in F$ such that $j' < j$, so $j' \in \text{Dom}(F)$. Suppose FTSOC $k' \geq k$. Then $J(j') \cong K(k')$. If $k' = k$, then $k \in \text{Rng}(F)$. Else, $k' > k$, so $K(k')(k) = K(k)$ is isomorphic to a segment of J , so $k \in \text{Rng}(F)$. Both cases are impossible, so $k' < k$.

Further, suppose there exist $(j', k') \in F$ such that $j' > j$. Then $J(j') \cong K(k')$. By similar logic, $J(j')(j)$ is isomorphic to a segment of K , so $j \in \text{Dom}(F)$, cont.

Hence, we have $F \subset J(j) \times K(k)$ is a bijection. It is easy to show that F is increasing, so $J(j) \cong K(k)$ and $(j, k) \in F$, cont.

If $\text{Dom}(F) = J$ and $\text{Rng}(F) = K$, then $J \cong K$ as F is increasing.

If $\text{Dom}(F) = J$ and $\text{Rng}(F) \subsetneq K$, pick $k \notin \text{Rng}(F)$ to be minimal. By similar logic, for all $k' \geq k$, $k' \notin \text{Rng}(F)$, so $F \subset J \times K(k)$ is order preserving, so $J \cong K(k)$.

Lastly, $\text{Dom}F \subsetneq J$ and $\text{Rng}F = K$. Similar to the previous case, $J(j) \cong K$ for some j .

Cases (1), (2), (3) are mutually exclusive. $\{(1), (2)\}, \{(1), (3)\}$ cannot happen since a well-ordered set cannot be isomorphic to a segment of itself. $\{(2), (3)\}$ is impossible: Suppose $J(j) \cong K$ and $J \cong K(k)$. Then there exists j' such that $J(j)(j') \cong K(k) \cong J$. $J(j)(j')$ is a segment of J .

2 Basic Ordinals

A set S is said to be transitive if for all $s \in S$, $s \subset S$.

S is transitive iff $\bigcup S \subset S$ and $S \subset P(S)$. If $t \in \bigcup S$, then $t \in s$ for some $s \in S$. Since $s \subset S$, we have $t \in S$. Conversely, if $\bigcup S \subset S$ and $s \in S$, then $s \subset \bigcup S \subset S$, so $s \subset S$. If S is transitive, $s \in S \rightarrow s \subset S \rightarrow s \in P(S)$, so $S \subset P(S)$. If $S \subset P(S)$ and $s \in S$, then $s \in P(S)$, or $s \subset S$.

A transitive set α whose elements are well-ordered by $\in$ is said to be an ordinal. Ex: $\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$ are ordinals. Denote by Ord the class of all ordinals.

If $\beta \in \alpha$ and $\alpha \in \text{Ord}$, then $\beta \in \text{Ord}$. Since $\beta \subset \alpha$, β is well-ordered by $\in$. Suppose $\gamma' \in \gamma \in \beta \in \alpha$. Since α is transitive, $\beta \subset \alpha$, so $\gamma \in \alpha$. Again, $\gamma \subset \alpha$, so $\gamma' \in \alpha$. $\gamma', \gamma, \beta \in \alpha$. Since the linear order of α wrt $\in$ is transitive, $\gamma' \in \beta$. Hence, $\gamma \subset \beta$, so β is transitive.

If $\alpha \in \text{Ord}$, then $\alpha^+ = \alpha \cup \{\alpha\} \in \text{Ord}$. If $\gamma \in \beta \in \alpha$, then $\gamma \in \alpha$ by transitivity of α , so the transitivity of $\in$ is preserved in α^+ . $\alpha \notin \alpha$ otherwise Regularity is contradicted. The minimal $\in$ element of any nonempty subset of α^+ still exists, so $\in$ is a well-order of α^+ .

$$\bigcup \alpha^+ = \bigcup \alpha \cup \{\alpha\} = \alpha \cup \bigcup \alpha = \alpha \subset \alpha^+$$

since $\bigcup \alpha \subset \alpha$ by transitivity of α . Hence, α^+ is transitive.

$\{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}\}$ is transitive but not well-ordered by $\in$. $\{\{\emptyset\}\}$ is well-ordered by $\in$ but not transitive.

If S, T are transitive, then so are $S \cap T$ and $S \cup T$. If $x \in S \cap T$, then $x \in S$, so $x \subset S$. Similarly, $x \subset T$, so $x \subset S \cap T$. Suppose $x \in S \cup T$. If $x \in S$, then $x \subset S \subset S \cup T$. If $x \in T$, then $x \subset T \subset S \cup T$, so $x \subset S \cup T$ either way. As a corollary, if $\alpha, \beta \in \text{Ord}$, then $\alpha \cap \beta \in \text{Ord}$. Since $\alpha \cap \beta \subset \alpha$ is well-ordered by $\in$ and $\alpha \cap \beta$ is transitive, $\alpha \cap \beta \in \text{Ord}$.

Suppose $\alpha, \beta \in \text{Ord}$ such that $\alpha \subsetneq \beta$. Then $\alpha \in \beta$. Let $\gamma \in \beta - \alpha$ be $\in$ -minimal. Suppose FTSOC there exists $\alpha' \in \alpha$ s.t. $\alpha' \geq \gamma$. If $\alpha' = \gamma$, then $\gamma \in \alpha$, cont. If $\gamma \in \alpha'$, since $\alpha' \subset \alpha$ as α is transitive, then $\gamma \in \alpha$, cont. Hence, $\alpha \subset \beta(\gamma) = \gamma$. If $\alpha \subsetneq \gamma$, then γ is not $\in$ -minimal, so $\alpha = \gamma \in \beta$.

Corollary, if $\alpha, \beta \in \text{Ord}$, then $\alpha = \beta$, $\alpha \in \beta$, or $\beta \in \alpha$. This is analogous to the statement for well-ordered sets. We have $\gamma = \alpha \cap \beta \in \text{Ord}$. If $\gamma \subsetneq \alpha$,

then $\gamma \in \alpha$. If $\gamma \subsetneq \beta$, then $\gamma \in \beta$. If $\gamma = \alpha$ and $\gamma = \beta$, then $\alpha = \beta$.

Corollary, if $\alpha, \beta \in \text{Ord}$, then $\alpha \cup \beta \in \text{Ord}$.

If $\mathcal{S}$ is a nonempty class of transitive sets S , then $\bigcap \mathcal{S}$ is transitive. If $x \in \bigcap \mathcal{S}$, then $x \in S$ for all S , so $x \subset S$ for all S , so $x \subset \bigcap \mathcal{S}$. Corollary: if $\mathcal{A}$ is a nonempty class of ordinals, $\bigcap \mathcal{A}$ is an ordinal.

Let $\mathcal{A}$ be a nonempty class of ordinals. Then $\alpha = \bigcap \mathcal{A} \in \mathcal{A}$ is an ordinal. Suppose FTSOC $\alpha \subsetneq \beta$ for all $\beta \in \mathcal{A}$. Then $\alpha \in \beta$ for all β , so $\alpha \in \alpha$, cont. We have $\alpha = \beta$ for some $\beta \in \mathcal{A}$. Denote α by $\min \mathcal{A}$.

Let $\mathcal{S}$ be a set of transitive sets. If $x \in \bigcup \mathcal{S}$, then $x \in S$ for some S , so $x \subset S$ for some S , so $x \subset \bigcup \mathcal{S}$, so $\bigcup \mathcal{S}$ is a transitive set.

If $\mathcal{A}$ is a set of ordinals, then $\sup \mathcal{A} = \bigcup \mathcal{A}$ is an ordinal. First, $\bigcup \mathcal{A}$ is a transitive set. Suppose $\alpha \in \beta \in \gamma$ s.t. $\alpha, \beta, \gamma \in \bigcup \mathcal{A}$. Then there exist $\alpha', \beta', \gamma' \in \mathcal{A} \subset \text{Ord}$ such that $\alpha \in \alpha', \beta \in \beta', \gamma \in \gamma'$. Then $\delta = \alpha' \cup \beta' \cup \gamma' \in \text{Ord}$. Since $\alpha, \beta, \gamma \in \delta$, then $\alpha \in \gamma$ by transitivity of $\in$ over δ . Since each $a \in \mathcal{A}$ is an ordinal, a is a set of ordinals. Hence, $\bigcup \mathcal{A}$ is also a set of ordinals. If $\emptyset \subsetneq S \subset \bigcup \mathcal{A}$, then $m = \min S = \bigcap S$ is an ordinal, so $\bigcup \mathcal{A}$ is well-ordered wrt $\in$.

Since α is a set of ordinals, $\bigcup \alpha$ is an ordinal. If $\alpha = \bigcup \alpha$, then α is said to be a limit ordinal.

If there exists a set α^- such that $\alpha = \alpha^- \cup \{\alpha^-\}$, then $\bigcup \alpha = \alpha^-$. Suppose for $\alpha, \beta \in \text{Ord}$ we have $\alpha \cup \{\alpha\} = \beta \cup \{\beta\}$. Then $\beta = \alpha$ or $\beta \in \alpha$. Suppose $\beta \in \alpha$ and $\alpha \in \beta$. Then by transitivity, $\alpha \subset \beta$ and $\beta \subset \alpha$, so $\alpha = \beta$. If α, β were arbitrary sets, by Regularity, $\alpha \in \beta, \beta \in \alpha$ is impossible. Either way, if the immediate predecessor α^- of α exists, α^- is unique.

Let $\mathcal{A}$ be a set of ordinals. Suppose $\sup \mathcal{A} \notin \mathcal{A}$. Then $\sup \mathcal{A}$ is a limit ordinal. (1) If $\alpha \in \mathcal{A}$, then $\alpha \subset \bigcup_{\alpha' \in \mathcal{A}} \alpha' = \bigcup \mathcal{A}$. (2) If $\exists \beta \in \text{Ord}$ st $\forall \alpha \in \mathcal{A}, \alpha \leq \beta$, then $\beta \geq \bigcup \mathcal{A}$. This follows from

$$\alpha \subset \beta \rightarrow \bigcup_{\alpha' \in \mathcal{A}} \alpha' \subset \beta \rightarrow \bigcup \mathcal{A} \subset \beta.$$

(3) The result. Let $\beta = \bigcup \mathcal{A}$. We wish to show $\beta \notin \mathcal{A} \rightarrow \bigcup \beta = \beta$. Suppose $\bigcup \beta \subsetneq \beta$. Then $\beta = \beta^- \cup \{\beta^-\}$ for some $\beta^- \in \text{Ord}$. Since for all $\alpha \in \mathcal{A}$, $\alpha \in \beta$, we have $\alpha \leq \beta^-$, so β^- is an upper bound for $\mathcal{A}$, cont min of β .

$\min\{\beta \in \text{Ord} : \beta > \alpha\} = \alpha^+$. If $\alpha \in \beta$, then $\alpha \subset \beta$, so $\beta \supset \alpha \cup \{\alpha\} = \alpha^+$. Since $\alpha^+ \in \text{Ord}$, we're done.

Ord is not a set. Sps FTSOC Ord is a set. If $\alpha, \beta \in \text{Ord}$, then α, β are comparable by $\in$. If $\alpha, \beta, \gamma \in \text{Ord}$ st $\alpha \in \beta \in \gamma$, then $\alpha \in \gamma$ since γ is transitive.

Hence, $\in$ is transitive over Ord . If $\emptyset \subsetneq \mathcal{A} \subset \text{Ord}$, then $\min \mathcal{A} \in \mathcal{A}$, so Ord is well-ordered by $\in$. Suppose $\alpha \in \bigcup \text{Ord}$. Then $\alpha \in \beta$ for some $\beta \in \text{Ord}$, so $\alpha \in \text{Ord}$. Hence Ord is transitive, so $\text{Ord} \in \text{Ord}$, cont Regularity.

Every well-ordered set is isomorphic to a unique ordinal. Let J be well-ordered. Append an artificial maximum element to J . Suppose there exists $j \in J$ st $J(j) \not\cong \alpha$ for all $\alpha \in \text{Ord}$. Let j be minimal, so for all $j' < j$ we have $J(j') \cong \alpha$ for some $\alpha \in \text{Ord}$. If j has an immediate predecessor j^- , we're done. If $J(j^-) \cong \alpha$, then $J(j) \cong \alpha^+$. Assume otherwise. α is unique: if $\alpha = \beta$, we're done. Else, WLOG $\alpha \in \beta$, so $\alpha = \beta(\alpha)$, cont. Hence, $F : J(j) \rightarrow \text{Ord}$ by $j' \mapsto \alpha$ is a class function.

By Replacement Schema, $\text{Rng}(F) = F(J(j))$ is a set, and in particular a set of ordinals. Let $\beta = \sup \text{Rng}(F)$. If $\beta \in \text{Rng}(F)$, then $\beta \cong J(j')$ for some $j' < j$. Since j has no predecessor, $j'^+ < j$ and $J(j'^+) \cong \beta^+$, contradicting the maximality of β . Hence, β is a limit ordinal and $\text{Rng}(F) \subset \beta$.

$F : J(j) \rightarrow \beta$ is increasing injective. Suppose F is not surjective, so for some $\gamma \in \beta$, $\gamma \notin \text{Rng}(F)$. Since β has no predecessor and $\beta \notin \text{Rng}(F)$, there exists $\gamma \in \gamma' \in \beta$ such that $\gamma' \in \text{Rng}(F)$. Since $\gamma' \cong J(j')$ and γ is a section of γ' , we have $\gamma \in \text{Rng}(F)$, cont. Hence, $F : J(j) \rightarrow \beta$ is an isomorphism, so $J(j) \cong \beta$, contradiction.

Take j to be the maximum element of J . Then our original well-ordered set $J(j)$ is isomorphic to some ordinal α .

Transfinite Induction. Let $\mathcal{C} \subset \text{Ord}$ be a class such that if $\alpha \in \mathcal{C}$, then $\alpha^+ \in \mathcal{C}$ and if α is a limit ordinal such that $\alpha \subset \mathcal{C}$, then $\alpha \in \mathcal{C}$. Then $\mathcal{C} = \text{Ord}$. Suppose $\text{FTSOC } \mathcal{C} \subsetneq \text{Ord}$. Pick $\alpha \in \text{Ord} - \mathcal{C}$ to be $\in$ -minimal. If α has a predecessor α^- , then $\alpha^- \in \mathcal{C}$, so $\alpha \in \mathcal{C}$, cont. Else, α is a limit ordinal. For all $\beta \in \alpha$, we have $\beta \in \mathcal{C}$. That is, $\alpha \subset \mathcal{C}$, so $\alpha \in \mathcal{C}$, cont.

Transfinite Recursion. Let C be a class and $\mathcal{F}$ denote the class of all class functions $F : \alpha \rightarrow C$ for some $\alpha \in \text{Ord}$. Let $G : \mathcal{F} \rightarrow C$ be a function. Then there exists a unique class function $F : \text{Ord} \rightarrow C$ such that $F(\alpha) = G(F|_\alpha)$ for all $\alpha \in \text{Ord}$.

First, we show there exists a unique $F : \alpha \rightarrow C$ for all $\alpha \in \text{Ord}$. If $\alpha = \emptyset$, then F is unique. Assume there exists a unique $F : \alpha \rightarrow C$ such that $F(\beta) = G(F|_\beta)$ for all $\beta \in \alpha$. Call this condition on F the G - F restriction. Let $F_1, F_2 : \alpha^+ \rightarrow C$ have the G - F restriction. Consider the function $F_1|_\alpha$. Let $\beta \in \alpha$.

$$(F_1|_\alpha)(\beta) = F_1(\beta) = G(F_1|_\beta) = G((F_1|_\alpha)|_\beta),$$

so $F_1|_\alpha$ has the G - F restriction and $F_1|_\alpha = F = F_2|_\alpha$. Since

$$F_1(\alpha) = G(F_1|_\alpha) = G(F) = G(F_2|_\alpha) = F_2(\alpha),$$

we have $F_1 \equiv F_2$. Such an F_1 exists in the first place since we can extend F from α to α^+ by taking $\alpha \mapsto G(F|\alpha)$.

Let α be a limit ordinal and suppose for all $\beta \in \alpha$, $F_\beta : \beta \rightarrow C$ is unique. If $\alpha' \in \alpha$, then $\alpha' + 1 \in \alpha$, so there exists $\alpha' \in \beta \in \alpha$. Define $F : \alpha \rightarrow C$ by $F(\alpha') = F_\beta(\alpha')$. This is well-defined, since if $\alpha' \in \beta, \beta' \in \alpha$, if $\beta = \beta'$, we're done. Else, WLOG $\beta \in \beta'$. It can be shown $F_{\beta'}|\beta$ satisfies the G - F restriction, so $F_{\beta'}|\beta = F_\beta$. Further,

$$F(\alpha') = F_\beta(\alpha') = G(F_\beta|\alpha') = G(F_{\alpha'}) = G(F|\alpha'),$$

so F satisfies the G - F restriction, showing existence. Now suppose there exist two such $F_1, F_2 : \alpha \rightarrow C$. Suppose FTSOC $F_1 \not\equiv F_2$, so $F_1(\beta) \neq F_2(\beta)$ for some $\beta \in \alpha$. Pick any $\beta \in \gamma \in \alpha$, so $F_1|\gamma, F_2|\gamma$ satisfy the G - F restriction, so $F_1|\gamma \equiv F_2|\gamma$, so

$$F_1(\beta) = (F_1|\gamma)(\beta) = (F_2|\gamma)(\beta) = F_2(\beta),$$

cont.

We extend to a class function with domain Ord. Let $F : \text{Ord} \rightarrow C$ by $F(\alpha) = F_\beta(\alpha)$ for some $\beta \ni \alpha$. F is well-defined since if $\beta \in \beta'$, then $F_{\beta'}|\beta = F_\beta$. F satisfies the G - F restriction. Uniqueness is similar to before.

If $\alpha \in \text{Ord}$ st $\alpha \neq \emptyset$, then $\emptyset \in \alpha$. $\emptyset \in \text{Ord}$, so $\alpha \in \emptyset$, $\alpha = \emptyset$, or $\emptyset \in \alpha$. Only the last case is possible. If α is a limit ordinal and $\beta \in \alpha$, then $\beta + 1 \in \alpha$. We have $\alpha \geq \beta + 1$. If $\alpha = \beta + 1$, then $\bigcup \alpha = \beta \neq \alpha$. Else $\beta + 1 < \alpha$ as desired. If $\alpha \neq \emptyset$ is an ordinal, then α is inductive.

3 The Integers

N is an ordinal. Let $N = \bigcap \{X : X \text{ is inductive}\}$. The axiom of infinity guarantees that N is well-defined. N is also inductive. Note that $N \cap \text{Ord} = N$. Suppose $y \in N$ is $\in$ -minimal such that $y \not\subset N$. $y \neq \emptyset$. Since N is the smallest inductive set, y is not inductive. Since y is nonzero noninductive ordinal, there exists $x \in \text{Ord}$ such that $y = x + 1$.

Since $x \cup \{x\} \not\subset N$, $x \not\subset N$ or $x \not\in N$. If $x \in N$ and $x \not\subset N$, then the $\in$ -minimality of y is contradicted. Then $x \not\in N$. Let $N' = N - \{y\}$. If $n \in N'$, then $n + 1 \in N$. If $n + 1 = y$, then $n = x$, but $x \not\in N$, so $n + 1 \in N'$. Further, $y \neq \emptyset$, so $\emptyset \in N'$, so N' is inductive, cont. N is transitive and well-ordered by $\in$, so $N \in \text{Ord}$. Following standard notation, we label N with ω for ordinals.

A set S is said to be finite if there exists a bijection between S and some

$n \in N$. Otherwise, S is infinite. N is infinite. Suppose $n \in N$ is $\in$ -minimal such that $n \stackrel{h}{\cong} N$. Note $n \neq \emptyset$ since $\emptyset \in N$. Hence, there exists $m \in N$ such that $n = m + 1$ (n is nonzero and not inductive). Suppose $h(m) = k$ for some $k \in N$. Then

$$m \cong N - \{k\} \cong N^\times \cong N,$$

so n is not $\in$ -minimal, cont.

If $n, m \in N$, then $n + m, n \cdot m \in N$. Let $I = \{m \in N : n + m \in N\}$. Note $0 \in I$ since $n + 0 = n \in N$. Suppose $m \in N$. Then $n + (m + 1) = (n + m) + 1 \in N$ since $n + m \in N$ and N is inductive. Thus, I is inductive, so $I = N$. The result follows.

Let $I = \{m \in N : n \cdot m \in N\}$. Note $n \cdot 0 = 0 \in N$ and $n \cdot (m + 1) = n \cdot m + n \in N$ (since $n \cdot m \in N$ and the sum of two integers is again an integer). Thus, $I = N$.

4 The Ordinals Again

If $\alpha \in \text{Ord}$, then $0 + \alpha = \alpha$. We proceed by transfinite induction. $0 + 0 = 0$. $0 + (\alpha + 1) = (0 + \alpha) + 1 = \alpha + 1$. $0 + \alpha = \sup\{0 + \beta : \beta < \alpha\} = \sup\{\beta : \beta < \alpha\} = \alpha$.

Let $\alpha, \beta, \gamma \in \text{Ord}$ st $\beta \leq \gamma$. Then $\alpha + \beta \leq \alpha + \gamma$ with equality iff $\beta = \gamma$ (so $+$ is strictly increasing in the second operand). (1) $\alpha + \beta = \alpha + \beta$. (2) $\alpha + (\gamma + 1) = (\alpha + \gamma) + 1 \geq \alpha + \gamma \geq \alpha + \beta$. (3) $\alpha + \gamma = \sup_{\gamma' < \gamma} \{\alpha + \gamma'\} \geq \sup_{\beta \leq \gamma' < \gamma} \{\alpha + \gamma'\} \geq \sup_{\beta \leq \gamma' < \gamma} \{\alpha + \beta\} = \alpha + \beta$. A strict inequality is achieved by applying this to $\alpha, \beta + 1$.

Denote by lOrd the class of limit ordinals. If $\alpha \in \text{Ord}$ and $\beta \in \text{lOrd}$, then $\alpha + \beta \in \text{lOrd}$. $\alpha + \beta = \sup_{\gamma < \beta} \{\alpha + \gamma\}$. If for some $\gamma < \beta$ the ordinal $\alpha + \gamma$ is maximal, then $\gamma + 1 < \beta$ and $\alpha + (\gamma + 1) = (\alpha + \gamma) + 1 > \alpha + \gamma$, cont max. Hence, $\{\alpha + \gamma : \gamma < \beta\}$ has no maximum, so $\alpha + \beta \in \text{lOrd}$.

For $\alpha \in \text{Ord}$, $1 + \alpha \geq \alpha$. We proceed by transfinite induction. (1) $1 + 0 = 1 \geq 0$. (2) $1 + (\alpha + 1) = (1 + \alpha) + 1 \geq \alpha + 1$. This follows from if $\alpha \leq \beta$, then $\alpha + 1 \leq \beta + 1$. If $\alpha = \beta$, we're done. Else, $\alpha \in \beta$ and note $\alpha \cup \{\alpha\} \subset \beta \cup \{\beta\}$. (3) $1 + \alpha = \sup_{\beta < \alpha} \{1 + \beta\} \geq \sup_{\beta < \alpha} \{\beta\} = \alpha$.

If $\mathcal{A} \subset \text{Ord}$ is a set and $x \in \text{Ord}$, then $x + \sup \mathcal{A} = \sup\{x + \alpha : \alpha \in \mathcal{A}\}$. If $\mathcal{A}$ has a maximum m , then $LHS = RHS = x + m$ since $+$ is increasing wrt the second operand. Else, (in general) $x + \sup \mathcal{A} \geq x + \alpha$ for all $\alpha \in \mathcal{A}$, so $x + \sup \mathcal{A} \geq \sup\{x + \alpha : \alpha \in \mathcal{A}\}$. Further, since $\mathcal{A}$ has no maximum,

$\sup \mathcal{A} \in \text{lOrd}$. We have

$$x + \sup \mathcal{A} = \sup_{\beta < \sup \mathcal{A}} \{x + \beta\} \leq \sup_{\alpha \in \mathcal{A}} \{x + \alpha\}.$$

The inequality follows from the fact that if for some $\beta < \sup \mathcal{A}$ we have for all $\alpha \leq \beta$, then $\sup \mathcal{A} \leq \beta < \sup \mathcal{A}$, cont. Hence, $x + \sup \mathcal{A} = \sup\{x + \alpha : \alpha \in \mathcal{A}\}$.

$+$ is associative over Ord. We proceed by transfinite induction. (1) $(\alpha + \beta) + 0 = \alpha + (\beta + 0)$. (2) $(\alpha + \beta) + (\gamma + 1) = ((\alpha + \beta) + \gamma) + 1 = (\alpha + (\beta + \gamma)) + 1 = \alpha + ((\beta + \gamma) + 1) = \alpha + (\beta + (\gamma + 1))$. (3) $(\alpha + \beta) + \gamma = \sup_{\gamma' < \gamma} \{(\alpha + \beta) + \gamma'\} = \sup_{\gamma' < \gamma} \{\alpha + (\beta + \gamma')\} = \alpha + \sup_{\gamma' < \gamma} \{\beta + \gamma'\} = \alpha + (\beta + \gamma)$.

$+$ is increasing in the first operand. Let $\alpha, \beta, \gamma \in \text{Ord}$. We wish to show if $\alpha \geq \beta$, then $\alpha + \gamma \geq \beta + \gamma$. We proceed by transfinite induction. (1) $\alpha + 0 \geq \beta + 0$. (2) $\alpha + (\gamma + 1) = (\alpha + \gamma) + 1 \geq (\beta + \gamma) + 1 = \beta + (\gamma + 1)$. (3) $\alpha + \gamma = \sup_{\gamma' < \gamma} \{\alpha + \gamma'\} \geq \sup_{\gamma' < \gamma} \{\beta + \gamma'\} = \beta + \gamma$.

Corollary: $1 + \omega = \sup_{n < \omega} \{1 + n\} \leq \sup_{n < \omega} \omega = \omega$. $1 + \omega \geq 0 + \omega = \omega$, so $1 + \omega = \omega$. $1 + \omega \neq \omega + 1$, so $+$ is not commutative over Ord.

Let $\alpha, \beta \in \text{Ord}$ st $\alpha \leq \beta$. Then there exists unique $\delta \in \text{Ord}$ st $\alpha + \delta = \beta$. $+$ is strictly increasing in the second operand, so ISTS the existence of δ . We proceed by transfinite induction on β . (1) $\alpha + 0 = \alpha$. (2) $\beta + 1 = (\alpha + \delta) + 1 = \alpha + (\delta + 1)$. (3) Let $\beta \in \text{lOrd}$ be st $\beta \geq \alpha$. Suppose for all $\beta' \in \text{Ord}$ st $\alpha \leq \beta' < \beta$ we have unique $\alpha + \delta' = \beta'$. Hence (by implicit application of axiom of Replacement, treating ‘ $-$ ’ as a function, denote by D the set of all such δ'). Then $\beta = \sup_{\beta' < \beta} \{\beta'\} = \sup_{\alpha \leq \beta' < \beta} \{\beta'\} = \sup_{\delta' \in D} \{\alpha + \delta'\} = \alpha + \sup D$.

$$\alpha \cdot 0 = 0 \cdot \alpha = 0. \quad 1 \cdot \alpha = \alpha \cdot 1 = \alpha.$$

$\cdot$ distributes over $+$. For $\alpha, \beta, \gamma \in \text{Ord}$, $\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$. We proceed by transfinite induction on γ . (1) $\alpha \cdot (\beta + 0) = \alpha \cdot \beta = \alpha \cdot \beta + \alpha \cdot 0$. (2) $\alpha \cdot (\beta + (\gamma + 1)) = \alpha \cdot ((\beta + \gamma) + 1) = \alpha \cdot (\beta + \gamma) + \alpha = (\alpha \cdot \beta + \alpha \cdot \gamma) + \alpha = \alpha \cdot \beta + (\alpha \cdot \gamma + \alpha) = \alpha \cdot \beta + \alpha \cdot (\gamma + 1)$. (3) Let $\gamma \in \text{lOrd}$. Then $\beta + \gamma \in \text{lOrd}$. $\alpha \cdot (\beta + \gamma) = \sup_{\delta < \beta + \gamma} \alpha \cdot \delta = \sup_{\beta \leq \delta < \beta + \gamma} \alpha \cdot \delta = \sup_{\gamma' < \gamma} \alpha \cdot (\beta + \gamma') = \sup_{\gamma' < \gamma} (\alpha \cdot \beta + \alpha \cdot \gamma') = \alpha \cdot \beta + \sup_{\gamma' < \gamma} \alpha \cdot \gamma' = \alpha \cdot \beta + \alpha \cdot \gamma$.

(It will be assumed that if $\mathcal{A} \subset \text{Ord}$ is a set and $x \in \text{Ord}$, then $x \cdot \sup \mathcal{A} = \sup\{x \cdot \alpha : \alpha \in \mathcal{A}\}$ (argument is similar to case $+$)).

$\cdot$ is associative. Let $\alpha, \beta, \gamma \in \text{Ord}$. Then $(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma)$. We proceed by transfinite induction on γ . (1) $(\alpha \cdot \beta) \cdot 0 = 0 = \alpha \cdot (\beta \cdot 0)$. (2) $(\alpha \cdot \beta) \cdot (\gamma + 1) = (\alpha \cdot \beta) \cdot \gamma + \alpha \cdot \beta = \alpha \cdot (\beta \cdot \gamma) + \alpha \cdot \beta = \alpha \cdot (\beta \cdot \gamma + \beta) = \alpha \cdot (\beta \cdot (\gamma + 1))$. (3) $(\alpha \cdot \beta) \cdot \gamma = \sup_{\gamma' < \gamma} (\alpha \cdot \beta) \cdot \gamma' = \sup_{\gamma' < \gamma} \alpha \cdot (\beta \cdot \gamma') = \alpha \cdot \sup_{\gamma' < \gamma} (\beta \cdot \gamma') = \alpha \cdot (\beta \cdot \gamma)$.

$$2 \cdot \omega = \sup_{n < \omega} 2 \cdot n \leq \sup_{n < \omega} \omega = \omega. \quad 2 \cdot \omega \geq 1 \cdot \omega = \omega, \text{ so } 2 \cdot \omega = \omega.$$

$\omega \cdot 2 = \omega + \omega \geq \omega + 1$, so $2 \cdot \omega \neq \omega \cdot 2$.

$(\omega + 1) \cdot \omega = \omega^2$. First, by induction, $(\omega + 1) \cdot n = \omega \cdot n + 1 \leq \omega \cdot (n + 1)$. Now $(\omega + 1) \cdot \omega = \sup_{n < \omega} (\omega + 1) \cdot n \leq \sup_{n < \omega} \omega \cdot (n + 1) = \sup_{n < \omega} \omega \cdot n = \omega^2$. Further, $(\omega + 1) \cdot \omega \geq \omega^2$, so $(\omega + 1) \cdot \omega = \omega^2$. $\omega + \omega^2 = \omega \cdot (1 + \omega) = \omega^2$.

Division. Let $a, d \in \text{Ord}$ and $d > 0$. There exist unique, $q, r \in \text{Ord}$ st $a = dq + r$. Let $Q \subset \text{Ord}$ be the class of all q st $dq \leq a$. Since $d > 0$, $d(a + 1) \geq a + 1 > a$, so $Q \subset a$ is a set. Since $d \sup Q = \sup\{dq : q \in Q\} \leq a$, $\sup Q \in Q$. Take q to be maximal.

Now since $dq \leq a$, there exists $r \in \text{Ord}$ fw $a = dq + r$. Sps FTSOC $r \geq d$. Then there exists $r' \in \text{Ord}$ fw $r = d + r'$. Then $a = dq + d + r' = d(q + 1) + r'$, cont max of q . Hence, $r < d$. Note subtraction when possible is unique, so given q , r is determined.

Suppose there exist $a = dq_1 + r_1 = dq_2 + r_2$. If q_1 is not maximal, $a \geq d(q_1 + 1) = dq_1 + d > dq_1 + r = a$. Hence, $q_1 = q_2 = q$. Then $r_1 = r_2 = r$.

Height of well-founded relation. Let E be a well-founded relation on P . There exists a unique function $\rho : P \rightarrow \text{Ord}$ st $\rho(x) = \sup\{\rho(y) + 1 : yEx\}$. Define inductively $P_0 = \emptyset$, $P_{\alpha+1} = \{x \in P : \forall y \in P(yEx \rightarrow y \in P_\alpha)\}$, and if $\alpha \in \text{lOrd}$, $P_\alpha = \bigcup_{\beta < \alpha} P_\beta$.

(1) I claim if $\alpha \leq \beta$, then $P_\alpha \subset P_\beta$. We induct on α . $\alpha = 0$ holds. Assume result for α , so $P_{\alpha+1} \subset P_{\beta+1}$, or that for all $\beta \geq \alpha + 1$ with $\beta \notin \text{lOrd}$, $P_{\alpha+1} \subset P_\beta$. If $\beta \in \text{lOrd}$, then $P_\beta = \bigcup_{\gamma < \beta} P_\gamma \supset P_{\alpha+1}$. If $\alpha \in \text{lOrd}$ and $\beta \geq \alpha$, $P_\alpha = \bigcup_{\gamma < \alpha} P_\gamma \subset \bigcup_{\gamma < \alpha} P_\beta = P_\beta$.

(2) There exists $\theta \in \text{lOrd}$ fw $P_\theta = P_{\theta+1}$. Suppose otherwise and consider the proper chain $\{P_\alpha\}_{\alpha \in \text{Ord}}$. Pick $x_\alpha \in P_{\alpha+1} - P_\alpha$. If $\alpha < \beta$, then $x_\beta \in P_{\beta+1} - P_\beta \subset P_{\beta+1} - P_{\alpha+1}$, but $x_\alpha \in P_{\alpha+1}$, so $x_\alpha \neq x_\beta$. We have an injection $\text{Ord} \hookrightarrow P$, which becomes a surjection $P \twoheadrightarrow \text{Ord}$, contradicting replacement.

(3) $P_\theta = P$. Assume $P_\theta \subsetneq P$ and let $x \in P - P_\theta$ be E -minimal. If yEx , then $y \in P_\theta$, so $x \in P_{\theta+1}$, so $P_\theta \neq P_{\theta+1}$. Take θ to be minimal (though this has no impact on the remainder of the proof).

(4) Define $\rho(x) = \min(\alpha \in \text{Ord} : x \in P_{\alpha+1})$. Let α be minimal st $x \in P_{\alpha+1}$. If yEx , then $y \in P_\alpha$ by construction of P . If $\alpha \notin \text{lOrd}$, then $y \in P_{(\alpha-1)+1}$, so $\rho(y) \leq \alpha - 1$, or that $\rho(y) + 1 \leq \alpha$. If $\alpha \in \text{lOrd}$, then for some $\gamma < \alpha$ we have $y \in P_\gamma \subset P_{\gamma+1} \subset P_\alpha$ (from $\gamma + 1 < \alpha$), so $\rho(y) \leq \gamma$ and $\rho(y) + 1 \leq \gamma + 1 < \alpha$. Hence, $\rho(x) \geq \sup\{\rho(y) + 1 : yEx\}$.

Now suppose $\rho(x) > \sup\{\rho(y) + 1 : yEx\} = \beta$. If yEx , then $y \in P_{\rho(y)+1} \subset P_\beta$, so $x \in P_{\beta+1}$, cont min of $\rho(x)$. Hence, $\rho(x) = \sup\{\rho(y) + 1 : yEx\}$.

(5) Uniqueness of ρ . Suppose there exist two ρ_1, ρ_2 and let x be E -minimal st $\rho_1(x) \neq \rho_2(x)$. For all yEx we have $\rho_1(y) = \rho_2(y)$, so

$$\rho_1(x) = \sup\{\rho_1(y) + 1 : yEx\} = \sup\{\rho_2(y) + 1 : yEx\} = \rho_2(x).$$

$0 < \gamma \in \text{lOrd}$ is said to be indecomposable if there doesn't exist $\alpha, \beta < \gamma$ st $\alpha + \beta = \gamma$. Show γ is indecomposable iff for all $\alpha < \gamma$, $\alpha + \gamma = \gamma$ iff $\gamma = \omega^\alpha$ for some α .

Suppose $\gamma > \alpha$. There exists δ fw $\gamma = \alpha + \delta$. If $\delta < \gamma$, cont., so $\delta \geq \gamma \rightarrow \delta = \gamma \rightarrow \alpha + \gamma = \gamma$. Conversely, suppose for all $\alpha + \gamma = \gamma$. Since $+$ is strictly increasing in the second operand, we can't have $\beta < \gamma$ and $\alpha + \beta = \gamma$.

Write in normal form $\gamma = \omega^{\beta_1}k_1 + \omega^{\beta_2}k_2 + \dots + \omega^{\beta_n}k_n$. If $nk_1 > 1$, γ is decomposable, so $\gamma = \omega^\beta$.

Claim ω^β is indecomposable for all $\beta \in \text{Ord}$. Holds for $\beta = 0, 1$. Assume result for β . Consider $\alpha < \omega^{\beta+1}$: $\alpha + \omega^{\beta+1} = \omega q + r + \omega^{\beta+1} = w(q + \omega^\beta) = ww^\beta = \omega^{\beta+1}$. Let $\beta \in \text{lOrd}$ and $\alpha < \omega^\beta$, so $\alpha < \omega^\delta$ for some $\delta < \beta$. $\omega^\delta + \omega^\beta = \sup_{\beta' < \beta} \{\omega^\delta + \omega^{\beta'}\} = \sup_{\delta < \beta' < \beta} \{\omega^\delta + \omega^{\beta'}\} = \sup_{\delta < \beta' < \beta} \{\omega^{\beta'}\} = \omega^\beta$.

5 The Principles

Well-Ordering principle implies Maximum principle. Suppose $(X, \leq)$ is a partial order. Let X be well-ordered, so index X by the ordinals contained in some $\gamma \in \text{Ord}$. Let $C_0 = \emptyset$ and define $C_{\alpha+1}$ to be $C_\alpha \cup \{x_\alpha\}$ if x_α is comparable to each element in C_α wrt $\leq$, otherwise $C_{\alpha+1} = C_\alpha$. The chains C_α inherit their ordering from X . If $\alpha \leq \beta$, then $C_\alpha \subset C_\beta$ (I should show this with transfinite induction to be formal).

If α is a limit ordinal, define $C_\alpha = \bigcup_{\beta < \alpha} C_\beta$ (implicitly $C_0 = \emptyset$ is a chain). Let $\leq_\beta \subset C_\beta^2$ be the linear order on C_β . Then take the order on C_α to be $\bigcup_{\beta < \alpha} \leq_\beta \subset C_\alpha^2$. If $x, y \in C_\alpha$, then there exist $\beta_1, \beta_2 < \alpha$ st $x \in C_{\beta_1}$, $y \in C_{\beta_2}$. Take $\beta = \max(\beta_1, \beta_2)$, so $x, y \in C_\beta$, so x, y are comparable in C_β and therefore C_α . If $x \leq_\alpha y$ and $y \leq_\alpha x$, then $x \leq_{\beta_1} y$ and $y \leq_{\beta_2} x$ for some $\beta_1, \beta_2 < \alpha$. $x \leq_\beta y$ and $y \leq_\beta x$, so $x = y$. $\leq_\alpha$ is antisymmetric. Now suppose $x \leq_{\beta_1} y$ and $y \leq_{\beta_2} z$. Then $x \leq_\beta y$ and $y \leq_\beta z$, so $x \leq_\beta z$, or $x \leq_\alpha z$. Hence, $\leq_\alpha$ is transitive, so $\leq_\alpha$ is a linear ordering (and C_α is indeed a chain).

I claim that the chain $C = C_\gamma$ is maximal. Suppose $x \notin C$ is comparable to every element in C . There exists $\alpha < \gamma$ such that $x = x_\alpha$. Since $C_\alpha \subset C_\gamma = C$,

x_α is comparable to every element in C_α . However, $x_\alpha \in C_{\alpha+1} \subset C_\gamma$, cont.

Maximum principle implies Zorn's lemma. Let $(X, \leq)$ be a partial order. Take $C \subset X$ to be a maximal chain. Let m be the maximum of C . If m is not maximal in X , there exists $m' \in X$ such that $m' > m$. Then m' is comparable to all elements of C (by transitivity) and $C \cup \{m'\} \supsetneq C$ is a chain, cont max.

Zorn's lemma implies any set has a linear order. Let $X \neq \emptyset$ be a set and let $Y = \{(\leq) \subset X^2 : \leq \text{ is a partial order of } X\}$. Note $\Delta X \in Y$, so $Y \neq \emptyset$. Let $\mathcal{C} \subset Y$ be a chain of partial orders on X . Then $\bigcup \mathcal{C}$ is also a partial order. By Zorn's lemma let $(\leq)$ be a maximal partial order. I claim $(\leq)$ is a linear order.

Suppose FTSOC there exist $(a, b), (b, a) \notin (\leq)$. Denote by $L_a = \{x \in X : (x, a) \in (\leq)\}$ and $R_b = \{x \in X : (b, x) \in (\leq)\}$. Let $(\leq') = (\leq) \cup (L_a \times R_b)$. I claim $(\leq')$ is a partial order.

First, $L_a \cap R_b = \emptyset$, otherwise for some $x \in X$, $(x, a) \in (\leq), (b, x) \in (\leq) \rightarrow (b, a) \in (\leq)$, cont. If $(x, y), (y, x) \in (\leq)$, then $x = y$ since $(\leq)$ is antisymmetric. If $(x, y) \in (\leq)$ and $(y, x) \in L_a \times R_b$, then $y \in L_a$, so $(y, a) \in (\leq) \rightarrow (x, a) \in (\leq) \rightarrow x \in L_a$. Since $x \in R_b$, this case isn't possible. If $(x, y), (y, x) \in L_a \times R_b$, then $x \in L_a \cap R_b = \emptyset$, cont. Hence, $(\leq')$ is antisymmetric.

Suppose $(x, y), (y, z) \in (\leq)$. Then $(x, z) \in (\leq) \subset (\leq')$. If $(x, y) \in (\leq)$ and $(y, z) \in L_a \times R_b$, then $x \in L_a$. Since $z \in R_b$, $(x, z) \in L_a \times R_b \subset (\leq')$. If $(x, y), (y, z) \in L_a \times R_b$, then $(x, z) \in L_a \times R_b \subset (\leq')$. Hence, $(\leq')$ is transitive.

$(\leq') \supsetneq (\leq)$ is a partial order, contradicting the maximality of $(\leq)$. Hence, $(\leq)$ is a linear order.

If $\alpha, \beta \in \text{Ord}$, then $\alpha\beta \sim \alpha \times \beta$ ($\sim$ is set isomorphism). Suppose $\gamma < \alpha\beta$. There exist unique $q, r \in \text{Ord}$ st $r < \alpha$ and $\gamma = \alpha q + r$. Let $f : \alpha\beta \rightarrow \alpha \times \beta$ be given by $\gamma \mapsto (r, q)$. Note f is well-defined injective (in particular $q < \beta$, else $\gamma = \alpha q + r \geq \alpha q \geq \alpha\beta > \gamma$). f is surjective. If $\beta \notin \text{lOrd}$, then $\gamma = \alpha q + r < \alpha(\beta - 1) + \alpha = \alpha\beta$. Else if $\beta \in \text{lOrd}$, then $\alpha q + r \leq \alpha q + \alpha = \alpha(q + 1) < \alpha\beta$ since $q < q + 1 < \beta$.

For $\alpha \in \text{Ord}$, $2^\alpha \geq \alpha$. We proceed by transfinite induction. (1) $2^0 = 1 \geq 0$, $2^1 = 2 \geq 1$. (2) $2^{\alpha+1} = 2^\alpha + 2^\alpha \geq \alpha + \alpha \geq \alpha + 1$. (3) Let $\alpha \in \text{lOrd}$. $2^\alpha = \sup_{\beta < \alpha} 2^\beta \geq \sup_{\beta < \alpha} \beta = \alpha$.

Corollary: $2^\omega = \sup_{n < \omega} 2^n \leq \sup_{n < \omega} \omega = \omega$. Hence, $2^\omega = \omega$ and there doesn't necessarily exist a bijection between α^β as an ordinal and α^β as a set product.

I believe $\alpha^{\beta+\gamma} = \alpha^\beta \alpha^\gamma$ and $(\alpha^\beta)^\gamma = \alpha^{\beta\gamma}$.

$$(\omega + 1) \cdot 2 = \omega + 1 + \omega + 1 = \omega + \omega + 1 = \omega \cdot 2 + 1 \neq \omega \cdot 2 + 2. \quad (\omega \cdot 2)^2 =$$

$$\omega \cdot 2 \cdot \omega \cdot 2 = \omega \cdot \omega \cdot 2 = \omega^2 \cdot 2 \neq \omega^2 \cdot 4.$$

Cantor's Normal Form. Any nonzero $\alpha \in \text{Ord}$ can be represented uniquely in the form

$$\alpha = \omega^{\beta_1} k_1 + \omega^{\beta_2} k_2 + \cdots + \omega^{\beta_n} k_n,$$

where $\alpha \geq \beta_1 > \beta_2 > \beta_3 > \cdots > \beta_n \geq 0$ and $0 < k_i < \omega$.

(1) Existence. Let $\alpha \in \text{Ord}$ be nonzero. Define $\mathcal{A} = \{\gamma \in \text{Ord} : \omega^\gamma \leq \alpha\}$. Note $\omega^{\alpha+1} \geq \alpha+1 > \alpha$, so $\mathcal{A}$ is bounded. Further, if $\gamma' < \gamma \in \mathcal{A}$, then $\omega^{\gamma'} < \omega^\gamma \leq \alpha$, so $\mathcal{A} \in \text{Ord}$ is transitive. If $\mathcal{A} \in \text{lOrd}$, then $\omega^{\mathcal{A}} = \sup_{\gamma \in \mathcal{A}} \omega^\gamma \leq \alpha$, so $\mathcal{A} \in \mathcal{A}$, cont. Let $\beta = \sup \mathcal{A} \in \mathcal{A}$.

Similarly, define $K = \{k \in \omega : \omega^\beta k \leq \alpha\}$. $K \in \text{Ord}$ is transitive and $K \neq 0$. $K \notin \text{lOrd}$ as otherwise $K = \omega$ (the minimal nonzero element of lOrd), for which $\omega^{\beta+1} = \omega^\beta \cdot \omega = \sup_{k < \omega} \omega^\beta k \leq \alpha$, cont max of β . Let $k = \sup K < \omega$ be nonzero.

Since $\alpha \geq \omega^\beta k$, there exists $\delta \in \text{Ord}$ such that $\alpha = \omega^\beta k + \delta$. Sps FTSOC $\delta \geq \alpha$, so $\alpha \geq \omega^\beta k + \alpha \geq \alpha \rightarrow \alpha = \omega^\beta k + \alpha$ (1). We can repeatedly replace α on the RHS of (1) with $\omega^\beta k + \alpha$, so by induction $\alpha = \omega^\beta k j + \alpha \geq \omega^\beta k j$ for all $j < \omega$. It follows $\alpha \geq \sup_{j < \omega} \omega^\beta k j = \omega^{\beta+1}$ cont max of β . Hence, we may assume $0 \leq \delta < \alpha$.

If $\delta = 0$, $\alpha = \omega^\beta k$ is in normal form. Else, write δ in normal form, so α is in almost normal form with $\beta_2 > \beta_3 > \cdots > \beta_n$, $0 < k_i < \omega$. If $\beta_1 < \beta_2$, then $\alpha \geq \omega^{\beta_2} k_2$, cont max of β_1 . If $\beta_1 = \beta_2$, then $\alpha \geq \omega^{\beta_1} (k_1 + k_2)$, cont max of k_1 , so $\beta_1 > \beta_2$, showing existence.

(2) Uniqueness. Suppose α is in normal form. Then

$$\begin{aligned} \alpha &\leq \omega^{\beta_1} k_1 + \omega^{\beta_2} k_2 + \omega^{\beta_2} k_3 + \cdots + \omega^{\beta_2} k_n \\ &= \omega^{\beta_1} k_1 + \omega^{\beta_2} (k_2 + k_3 + \cdots + k_n) \\ &< \omega^{\beta_1} k_1 + \omega^{\beta_2+1} \\ &\leq \omega^{\beta_1} k_1 + \omega^{\beta_1} \\ &= \omega^{\beta_1} (k_1 + 1), \end{aligned}$$

so β_1, k_1 are both maximal (and therefore unique). In principle this process can be repeated or formalized with an induction argument.